

Formula Sheet

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$E(X) = \mu = \sum_{all\ x} x f(x)$$

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = \frac{\sum_{i=1}^n x_i^2 - n\bar{x}^2}{n-1}$$

$$V(X) = E(X^2) - [E(X)]^2$$

$$\binom{n}{x} p^x (1-p)^{n-x}$$

$$\frac{e^{-\lambda} \lambda^x}{x!}$$

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$\frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$\frac{\bar{x}_D - \mu_D}{s_D/\sqrt{n_D}}$$

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

$$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)}}, \quad \nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

$$\sum_{i=1}^k \frac{(f_i - e_i)^2}{e_i}$$

Confidence Interval: estimate $\pm (c.v.)(e.s.e.)$ or estimate $\pm (c.v.)(s.e.)$

Regression Formula Sheet

Data: $(x_i, y_i), \quad i = 1, 2, \dots, n$

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right]$$

$$s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 = \frac{1}{n-1} \left[\sum_{i=1}^n y_i^2 - \frac{1}{n} \left(\sum_{i=1}^n y_i \right)^2 \right]$$

$$s_{xy} = \frac{1}{n-1} \left(\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right) = \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \frac{1}{n} \sum_{i=1}^n x_i \sum_{i=1}^n y_i \right]$$

$$\hat{y} = b_0 + b_1 x \qquad b_1 = \frac{s_{xy}}{s_x^2} \qquad b_0 = \bar{y} - b_1 \bar{x}$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = (n-1) \left(s_y^2 - \frac{s_{xy}^2}{s_x^2} \right)$$

$$s_\varepsilon = \sqrt{\frac{SSE}{n-2}}$$

$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

$$s_{b_1} = \frac{s_\varepsilon}{\sqrt{(n-1)s_x^2}}$$

$$R^2 = \frac{s_{xy}^2}{s_x^2 s_y^2}$$

$$r = \frac{s_{xy}}{s_x s_y}$$

$$t = r \sqrt{\frac{n-2}{1-r^2}}$$